From Instruction to Interaction: Designing an Embodied, AI-Supported Learning Ecology

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ABSTRACT

Despite advances in artificial intelligence (AI) in education, classroom practice remains largely instructionist, emphasizing delivery and measurement over interaction, exploration, and meaning-making. Building on embodied cognition, sociocultural theory, and postdigital perspectives, my poster proposes an embodied, AI-supported learning ecology that integrates extended reality (XR), sensing technologies, and AI. The ecology aims to treat movement, gesture, and affect as pedagogically meaningful data, enhancing learner engagement and supporting more responsive, human-centred teaching.

INTRODUCTION

Learning is inherently situated, embodied, and socially mediated: people think with and through their bodies, tools, and environments, not in isolation. Yet many AI-driven systems in education still prioritize individualized content delivery and performance metrics, reinforcing instructionist logics and narrowing what counts as participation. This work introduces an embodied, AI-supported learning ecology that foregrounds interaction, movement, and context, seeking to better align AI-mediated teaching with how learners experience the world.

CONCEPTUAL FRAMEWORK

My project is grounded in three intertwined strands of scholarship:

- Embodied and grounded cognition. Work in psychology and the learning sciences argues that cognitive processes are deeply grounded in perception and action systems, and documents links between bodily action, affect, and conceptual understanding in domains such as mathematics and physics (e.g., Barsalou, 2008; Wilson, 2002; Alibali & Nathan, 2012).

- Sociocultural and situated learning. Learning is understood as participation in culturally and materially organized activities, where tools, artifacts, and other people's actions shape what can be seen and done (e.g., Vygotsky, 1978; Lave & Wenger, 1991; Dewey, 1938; Freire, 1970).

- Postdigital and critical perspectives on AI. Contemporary educational systems unfold across platforms, data infrastructures, and institutional practices, raising questions about power, surveillance, and the purposes of education (e.g., Biesta, 2010, 2022; Fawns, 2019; Jandrić, 2019; Knox, 2020; Zuboff, 2019).

The proposed learning ecology integrates XR environments, wearable sensing, and AI systems to make movement, gesture, and affect pedagogically meaningful. Wearables and environmental beacons capture multimodal traces of learners' activity; AI models analyze these traces in relation to XR tasks and generate real-time insights for both teachers and learners.
Illustrative scenario: Consider a high-school student walking home after physics class. As part of the XR–AI ecology, she puts on AR glasses that invite her to explore acceleration along her everyday route. The system asks her to gradually increase and decrease her walking speed; as she does so, a virtual velocity–time graph floats in her field of view, dynamically updating slope and values in response to her movement. She feels the bodily sensation of speeding up and slowing down while seeing how these sensations map onto graphical representations and formal definitions of acceleration. This scenario extrapolates from research on embodied and mixed-reality learning, which

suggests that coupling bodily experience with dynamic representations can support durable understanding of disciplinary ideas (e.g., Abrahamson & Sánchez-García, 2016; Lindgren & Johnson-Glenberg, 2013; Johnson-Glenberg et al., 2014).

RESEARCH QUESTIONS

Guided by a design-based research stance, my work asks:

- Engagement: How can an embodied, AI-supported learning ecology enhance students' behavioral, emotional, and cognitive engagement in disciplinary learning across XR-mediated units?
- Teacher agency: How can AI-mediated feedback support teacher noticing, interpretation, and action while preserving professional judgment and student autonomy?
- Ethics and experience: How do students and teachers experience being part of an ecology in which bodily activity and everyday movement are sensed, modeled, and fed back as pedagogical resources?

DESIGN OVERVIEW

The initial design targets high-school mathematics and physics topics that lend themselves to spatial and bodily exploration (e.g., vectors, functions, acceleration, forces).

XR tasks: Learners engage in mixed-reality activities such as walking coordinate planes, embodying vectors, or exploring acceleration through everyday movement scenarios like the AR walk described above, building on prior work in embodied and mixed-reality STEM learning (e.g., Abrahamson & Sánchez-García, 2016; Lindgren & Johnson-Glenberg, 2013; Johnson-Glenberg et al., 2014).

Sensing infrastructure: Wearable devices (e.g., smartwatches, AR headsets), environmental beacons, and mobile sensors collect data on position, motion, gesture, and basic affective indicators (e.g., heart-rate variability).

AI analytics: Models fuse these multimodal traces with task context to infer indicators of engagement and productive struggle, generating low-latency visualizations and prompts—for example, highlighting groups whose movement patterns suggest confusion, or inviting learners to re-enact a motion in slow motion to reflect on a concept.

Teacher dashboard: A dashboard displays evolving indicators of class-wide and small-group engagement, offers suggested prompts or regroupings, and allows teachers to annotate or override AI suggestions. The aim is to augment, not automate, pedagogical decision-making, echoing human-centred AI principles and recent work on AI ethics in society and education (Floridi & Cowls, 2019; Williamson & Eynon, 2020; Knox, 2020).

Ecological scope: Activities extend beyond the classroom into home and community spaces: walking routes, playgrounds, or local transit can become sites for embodied inquiry, blurring boundaries between "school learning" and everyday experience (Dewey, 1938; Taylor, 2017; Fawns, 2019).

VALIDATION PLAN

A multi-cycle design-based research program will guide the development:

Context and participants: Pilot implementations in two public high schools, in collaboration with mathematics and physics teachers serving diverse student populations.

Design cycles: Co-design workshops with teachers and students will shape XR tasks, sensing configurations, and dashboard features, followed by classroom trials and reflective redesign.

Data sources:

- Multimodal traces: sensor data, interaction logs, and XR event streams.
- Video and fieldnotes documenting classroom activity and teacher moves.
- Interviews and focus groups with students and teachers about experiences of embodiment, agency, and AI mediation.
- Short surveys grounded in self-determination theory to probe perceived autonomy, competence, and relatedness (Deci & Ryan, 1985, 2000; Niemiec & Ryan, 2009).

Analytic focus: Analyses will examine how embodied activity, AI-generated feedback, and teacher decisions intertwine, and how students narrate their learning experiences across school and everyday settings. The goal is to derive transferable design principles and ethical guidelines for embodied, AI-supported learning ecologies.

CONTRIBUTIONS

This project aims to contribute:

A conceptual framework for embodied, AI-supported learning ecologies that bridges embodied cognition, sociocultural theory, and postdigital critiques of data-driven education (e.g., Barsalou, 2008; Wilson, 2002; Biesta, 2010, 2022; Jandrić, 2019; Zuboff, 2019).

A system-level design integrating XR, sensing, and AI to treat bodily activity and context as a first-class learning resource.

Design principles for leveraging embodied data to foster engagement, autonomy, and relatedness, while respecting concerns about surveillance, measurement, and professional expertise (e.g., Biesta, 2010; Fawns, 2019; Floridi & Cowls, 2019; Williamson & Eynon, 2020).

Implications for education futures, contributing to debates about the roles of AI and automation in professional work and schooling (e.g., Frey & Osborne, 2017; Papert, 1980; Susskind & Susskind, 2015; Knox, 2020).

CONCLUSION

By shifting focus from instruction to interaction, the proposed learning ecology reframes learning as an embodied, world-centred, and socially situated process rather than a sequence of decontextualized content deliveries (Biesta, 2010, 2022; Dewey, 1938; Lave & Wenger, 1991). XR, sensing, and AI are treated not as neutral delivery channels but as partners in re-designing how concepts are encountered in everyday life, how bodies participate in disciplinary practice, and how teachers exercise judgment in data-rich environments. The design-based research project seeks to advance human-centred AI in education by showing how technologies can be oriented toward richer forms of experience, agency, and shared inquiry rather than mere efficiency and control.

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